

Q1

i) Transport layer (between processes) connects the sender and receiver (applications) by having all the data in segments and deliver those segments using TCP/UDP. Hence, it provides logical communication due to above rather than physical communication.

ii) UDP being fast and not having connection made between sender and receiver is preferred in online gaming, video streaming, and live broadcasts as well etc.

iii) `rdt-rcv()` is called when packets coming from the channel to the receiver's side, then you need to check if either to deliver the data or send ACK/NAK.

`rdt-send()` is called by application layer when it needs to send data, where packet is created and then sequence no.# and checksum applied.

iv) It is used to transmit packet from sender side to the network, done after `rdt-send` but since this data transfer is unreliable so the channel may corrupt or lose packets.

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- v) There are two planes, control and data plane. Control plane makes sure where to send packet and then data plane which actually forwards that packet. So SDN gives the ability to program the control plane alone through software due to which network management can be easily done and separates control from data plane.
- vi) Traditional network has a static control plane which can be configured manually, and is distributed means for each device. However, SDN has centralized control plane which is programmable by the controller and not to be configured at each device, so much flexible.

Q2

20 bytes used in header, so 1480 bytes of data.
no. of datagrams = $\frac{48840}{1480} = 33$ (if answer was in

in decimal, then 1 more datagram would have been needed).

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Q3

224 bits = 28 bytes , 8 bits = 1 bytes

23K-2001 , $x = 02$, $y = 0$, $z = 0$

1. Source port number = 0
2. Destination port number = 0
3. sequence number = 1
4. acknowledgement number = 0
5. TCP header length (data-offset field) = 3
header size = $3 \times 4 = 12$ bytes
6. ACK bit = 1
7. FIN bit = 0
8. SYN bit = 0
9. Window size = 0
10. checksum = 2

0000 0000 0000 0001 0000 0000

3FF0 0000 0002 FFFF 0100 0000 0010 0000